The Journal of Immunology

RESEARCH ARTICLE | MAY 17 2023

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J Immunol (2023) 211 (1): 71–80.
https://doi.org/10.4049/jimmunol.2300122

Spi-C and PU.1 Counterregulate *Rag1* and *Igκ* Transcription to Effect Vκ-Jκ Recombination in Small Pre-B Cells

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B cell development requires the ordered rearrangement of *Ig* genes encoding H and L chain proteins that assemble into BCRs or Abs capable of recognizing specific Ags. *Igκ* rearrangement is promoted by chromatin accessibility and by relative abundance of RAG1/2 proteins. Expression of the E26 transformation-specific transcription factor Spi-C is activated in response to dsDNA double-stranded breaks in small pre-B cells to negatively regulate pre-BCR signaling and *Igκ* rearrangement. However, it is not clear if Spi-C regulates *Igκ* rearrangement through transcription or by controlling RAG expression. In this study, we investigated the mechanism of Spi-C negative regulation of *Igκ* L chain rearrangement. Using an inducible expression system in a pre-B cell line, we found that Spi-C negatively regulated *Igκ* rearrangement, *Igκ* transcript levels, and *Rag1* transcript levels. We found that *Igκ* and *Rag1* transcript levels were increased in small pre-B cells from *Spic*$^{-/-}$ mice. In contrast, *Igκ* and *Rag1* transcript levels were activated by PU.1 and were decreased in small pre-B cells from PU.1-deficient mice. Using chromatin immunoprecipitation analysis, we identified an interaction site for PU.1 and Spi-C located in the *Rag1* promoter region. These results suggest that Spi-C and PU.1 counterregulate *Igκ* transcription and *Rag1* transcription to effect *Igκ* recombination in small pre-B cells. ***The Journal of Immunology*, 2023, 211: 71–80.**

B cell development requires the ordered rearrangement of *Ig* genes encoding H and L chain proteins that assemble into BCRs or Abs capable of recognizing specific Ags. *Ig* rearrangement proceeds in a developmental sequence in which *IgH* V(*D*)J recombination precedes Igκ or Igλ V-J recombination. Igκ/λ V-J recombination occurs in small pre-B cells that have previously rearranged a functional *IgH* allele, have ceased proliferation in response to IL-7, and have turned on high expression of recombinase-activating gene (RAG) proteins RAG1 and RAG2 (1, 2). *Igκ* rearrangement is promoted by chromatin accessibility, regulated through transcription, and regulated by relative abundance of RAG1/2 proteins (3).

Igκ accessibility and transcription are controlled by two locus-specific enhancers: the intronic enhancer and the 3′ enhancer (4). *Igκ* chromatin accessibility correlates with germline transcription initiating upstream of the Jκ1 segment; this transcript is alternatively named k^0 or germline κ-1 (*Glk1*) (5, 6). The 3′ enhancer contains an E26 transformation-specific (ETS) transcription factor binding site that regulates B cell versus T cell specificity of Vκ-Jκ joining (7). *Igκ* transcription and accessibility are repressed by STAT5 interaction with the *Igκ* intronic enhancer downstream of IL-7 receptor signaling in pro-B cells and large pre-B cells (8, 9).

Regulation of RAG protein abundance is an important second mechanism for regulation of *Igκ* recombination. RAG1 and RAG2 together form the recombinase that catalyzes V(*D*)J recombination by recognition and cleavage of recombination signal sequences (10). Appropriate developmental regulation of *Rag1*/*Rag2* is essential to prevent off-target mutations leading to leukemia or lymphoma (11). RAG1/2 protein levels are regulated by specific degradation linked to the cell cycle (12). The closely linked *Rag1* and *Rag2* genes are also regulated transcriptionally (13). Several regulatory regions or enhancers of the *Rag* locus have been identified, including Erag (14), R-Ten, and R1B/R2B (15). These regulatory regions interact with lineage-specific transcription factors to activate or repress *Rag* transcription at specific developmental stages (15). RAG-induced DNA double-stranded breaks (DSBs) promote allelic exclusion by signaling through ataxia telangiectasia mutated (ATM) kinase and the Nemo (NF-κB essential modulator) protein to repress *Rag* and *Igκ* transcription (16).

One of the genes activated by DSBs in small pre-B cells through the ATM/NF-κB signaling pathway is *Spic* encoding the ETS transcription factor Spi-C. Spi-C is a lineage-instructive transcription factor that is important for the generation of myeloid and lymphoid cells (17). Induction of Spi-C in pre-B cells functions to repress cell cycle progression and *Igκ* recombination by interacting with and repressing the *Syk*, *Blnk*, and *Igκ* genes (18, 19). Spi-C is required for the development of splenic red pulp macrophages (20, 21). Spi-C is dynamically regulated in response to growth factor signals to regulate gene expression in B cells (22, 23).

Spi-C is closely related to the ETS family transcription factors PU.1 and Spi-B that function as complementary transcriptional activators of BCR signaling and *Igκ* rearrangement (24). Combined deletion of *Spi1* and *Spib* genes encoding PU.1 and Spi-B in mice results in a block to B cell development at the small pre-B cell stage followed by development of precursor B cell acute lymphoblastic leukemia (25, 26). PU.1 and Spi-B

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Received for publication February 17, 2023. Accepted for publication May 2, 2023.

This work was supported by the Natural Sciences and Engineering Research Council of Canada Discovery Grant 2022-03518 and Canadian Institutes of Health Research Operating Grant 168995 (R.P.D.).

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The online version of this article contains supplemental material.

Abbreviations used in this article: Abl, Abelson; ATM, ataxia telangiectasia mutated; BM, bone marrow; ChIP, chromatin immunoprecipitation; DSB, double-stranded break; ETS, E26 transformation-specific; RAG, recombinase-activating gene; RT-qPCR, quantitative RT-PCR.

www.jimmunol.org/cgi/doi/10.4049/jimmunol.2300122

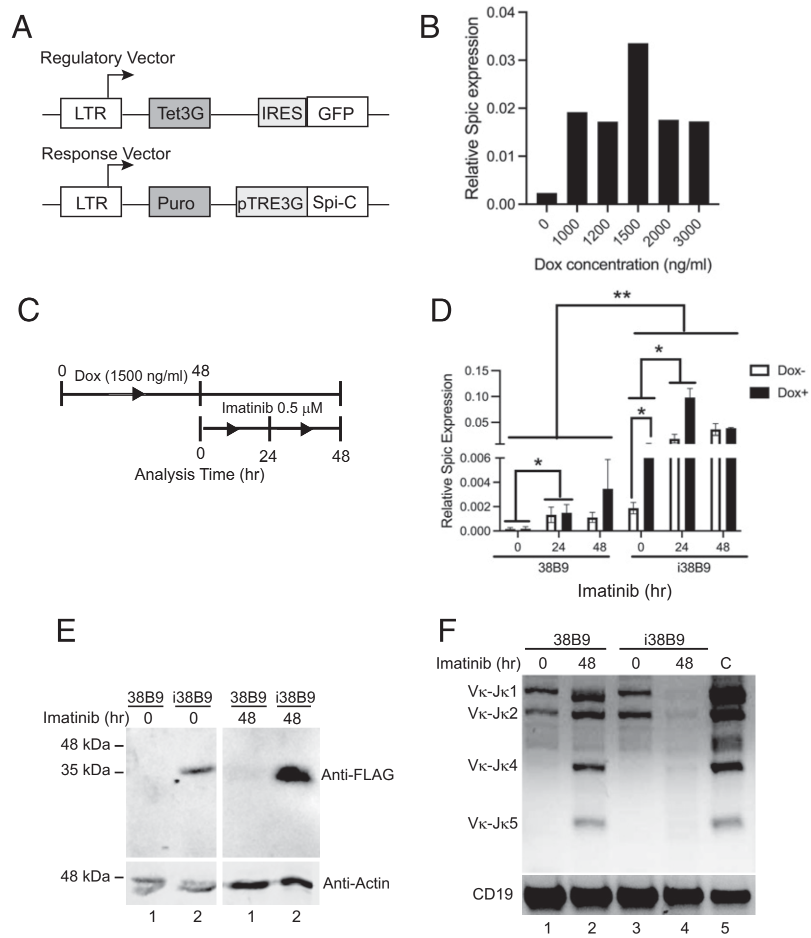

FIGURE 1. *Igκ* rearrangement is inhibited by Spi-C in an inducible cell line. (**A**) Schematic of two-vector inducible system. The top panel shows the regulatory vector, and the bottom panel shows the response vector. LTR, long terminal repeat; Tet3G, third-generation tetracycline-regulatable activator; IRES, internal ribosomal entry site; Puro, puromycin resistance gene; pTRE3G, third-generation tetracycline regulatable element. (**B**) Induction of *Spic* expression by doxycycline in inducible 38B9 (i38B9) cells. Doxycycline (Dox) concentrations in ng/ml are indicated on the *x*-axis, Relative *Spic* expression (*Spic*/β-*actin* ratios) are indicated on the *y*-axis. (**C**) Time course of induction of *Spic* expression by doxycycline and differentiation by imatinib in i38B9 cells. (**D**) Induction of *Spic* expression by doxycycline and imatinib in i38B9 cells. Times after treatment with 0.5 μM imatinib are shown on the *x*-axis. Induction with 1.5 μg/ml doxycycline is indicated by open or filled bars. Relative *Spic* expression (*Spic*/β-*actin* ratios) is indicated on the *y*-axis. (**E**) Immunoblot analysis of Spi-C expression. Lysates prepared from 38B9 or i38B9 cells induced with 1.5 μg/ml doxycycline, then untreated (0) or treated for 48 h (48) with imatinib, were probed with anti-FLAG Ab to detect 3xFLAG-tagged Spi-C protein (upper panels) or anti-β-actin (lower panels). (**F**) Inhibition of *Igκ* V-J rearrangement 48 h after imatinib treatment and Spi-C induction. PCR analysis was performed on genomic DNA prepared from cells induced with 1.5 μg/ml doxycycline, then untreated (0) or treated for 48 h (48) with imatinib. Positive control (C) is from genomic DNA prepared from wild-type splenic B cells. Lower panel shows products of PCR amplification of *Cd19*. $^{*}p < 0.05$ and $^{**}p < 0.01$ using one-way ANOVA.

function as transcriptional activators, whereas Spi-C functions as a transcriptional repressor by competing for PU.1 binding sites (23, 27, 28). In addition, DSBs induced by RAG activate Spi-C to displace PU.1 binding from the *Igκ* 3′ enhancer to repress transcription (19). Therefore, PU.1/Spi-B and Spi-C may function to oppose one another's function to regulate BCR signaling and *Igκ* rearrangement during B cell development.

In this study, we investigated the hypothesis that Spi-C and PU.1 function as regulators of *Igκ* recombination by regulation of *Rag* transcription as well as *Igκ* transcription. To do this, we constructed a doxycycline-inducible Spi-C expression system in 38B9 pre-B cells. Induction of Spi-C inhibited *Igκ* recombination and inhibited *Igκ* and *Rag* mRNA transcription in 38B9 cells. To determine if Spi-C is required to repress *Igκ* and *Rag* mRNA transcription during B cell

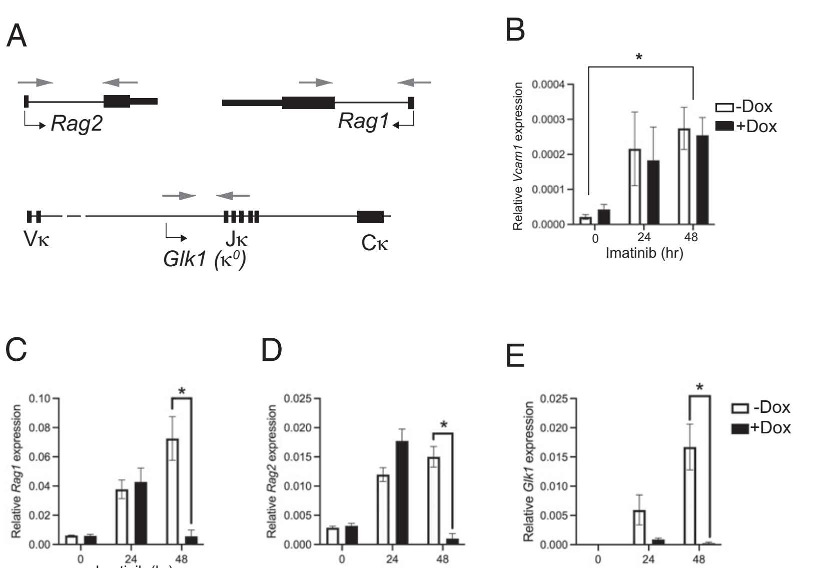

FIGURE 2. Repression of *Igκ* germline and *Rag1* transcripts by Spi-C. RNA was prepared from i38B9 cells treated for 0, 24, or 48 h with 0.5 μM imatinib and induced or not with 1.5 mg/ml doxycycline (open versus filled bars). (**A**) Primer locations for *Rag1*, *Rag2*, or germline κ (*Glk1*) are indicated with light gray arrows. Relative expression (gene/β-*actin* ratios) was determined for (**B**) *Vcam1*, (**C**) *Rag1*, (**D**) *Rag2*, or (**E**) *Glk1*. $^{*}p < 0.05$ using one-way ANOVA and Dunnett's multiple comparisons test.

Table I. Expected and actual frequencies of *Spic*$^{-/-}$ mice at 14.5 d after implantation or 3 wk after birth from F2 *Spic*$^{+/-}$ intercrosses

F1 Genetic Background	Mouse Age upon Genotyping Characterization	Genotype	No. of Mice Produced	Expected Frequency[a] (%)	Actual Frequency[b] (%)
		Spic$^{+/+}$	6	25	15.4
C57BL/6	14.5 d postimplantation	Spic$^{+/-}$	31	50	79.5
		Spic$^{-/-}$	2	25	5
		Spic$^{+/+}$	5	25	21.7
C57BL/6	3 wk	Spic$^{+/-}$	18	50	78.3
		Spic$^{-/-}$	0	25	0
		Spic$^{+/+}$	99	25	30
C57BL and 129	3 wk	Spic$^{+/-}$	182	50	55.2
		Spic$^{-/-}$	49	25	14.8

[a]Based on Mendelian ratio.
[b]Based on the calculation from total mice within the group.

development, we generated *Spic*$^{-/-}$ mice by intercrossing *Spic*$^{+/-}$ mice onto a mixed C57BL/6 and 129S/v background. *Ig*κ and *Rag* mRNA transcripts were upregulated in small pre-B cells enriched from *Spic*$^{-/-}$ mice relative to wild-type pre-B cells. *Ig*κ and *Rag1* transcript levels were activated by PU.1. Finally, we used chromatin immunoprecipitation (ChIP) to identify a regulatory binding site for PU.1 and Spi-B upstream of the *Rag1* gene. These results indicate that Spi-C and PU.1 counterregulate *Rag* and *Ig*κ transcription during B cell development.

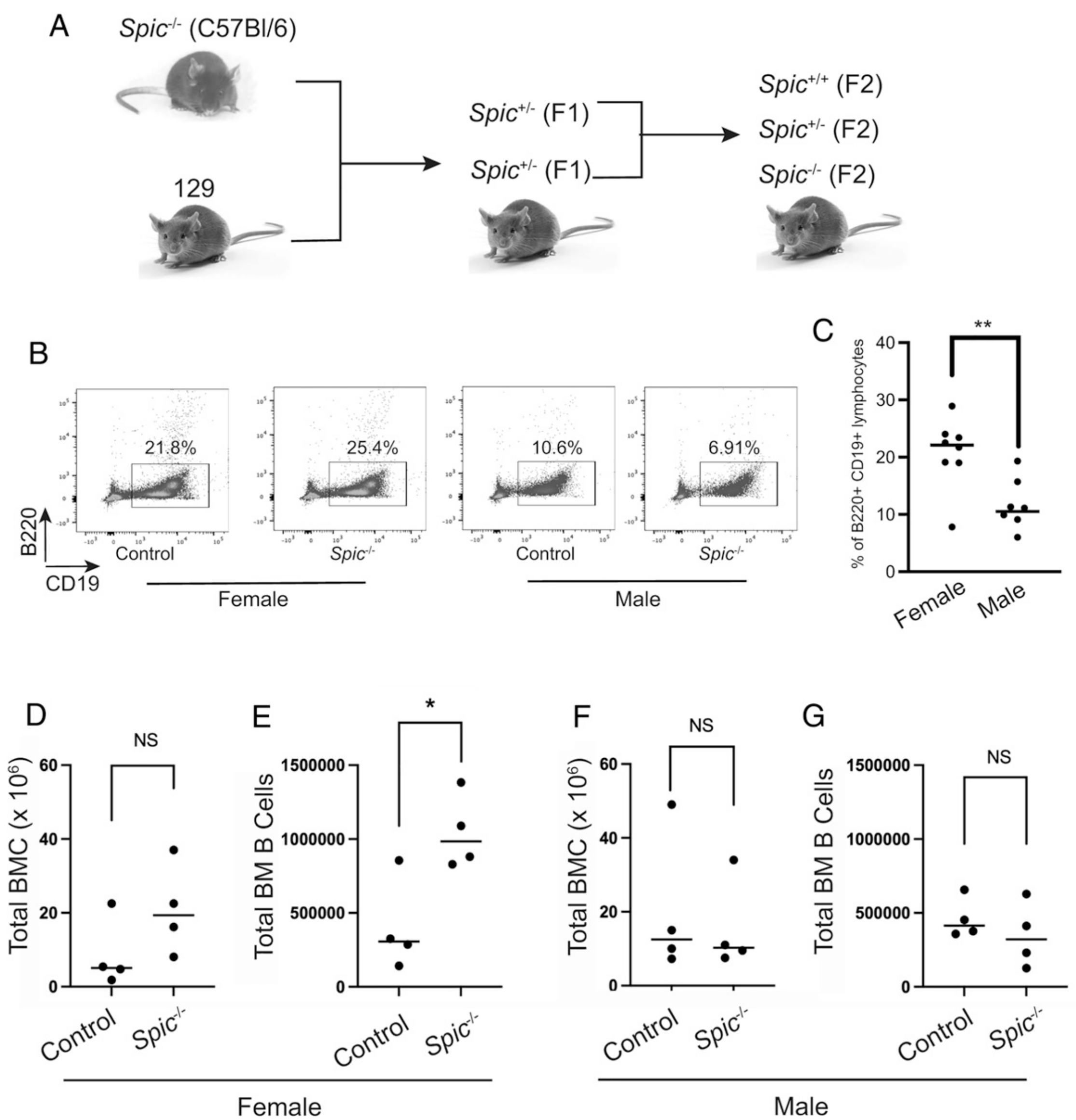

FIGURE 3. Generation of *Spic*$^{-/-}$ mice. (**A**) Schematic of breeding strategy. (**B**) Representative flow cytometric analysis of B220^{+} CD19^{+} B cell frequencies in the BM of mice of the indicated genotypes. (**C**) Frequencies of B220^{+} CD19^{+} B cells in the BM of female and male F2 mice. $^{**}p < 0.01$ using Student *t* test. (**D**) Absolute (total) numbers of erythrocyte-depleted BM cells in female control and *Spic*$^{-/-}$ F2 mice. Control mice include *Spic*$^{+/+}$ and *Spic*$^{+/-}$ mice. (**E**) Absolute numbers of B220^{+} CD19^{+} B cells in the BM of female control and *Spic*$^{-/-}$ F2 mice. $^{*}p < 0.05$ using Student *t* test. (**F**) Absolute numbers of BM cells in male control and *Spic*$^{-/-}$ F2 mice. (**G**) Absolute numbers of B220^{+} CD19^{+} B cells in the BM of male control and *Spic*$^{-/-}$ F2 mice. NS, not significant.

Materials and Methods

Mice

Spic$^{+/-}$ mice were maintained as previously described on a C57BL/6 background (29). To generate *Spic*$^{-/-}$ mice, *Spic*$^{+/-}$ mice were mated to 129Sv mice (Charles River Laboratories, Pointe-Claire, QC, Canada) to generate F1 C57BL/6/129Sv *Spic*$^{+/-}$ mice. F1 mice were intercrossed to generate F2 *Spic*$^{-/-}$ mice. All experiments with F2 *Spic*$^{-/-}$ mice used F2 *Spic*$^{+/+}$ or *Spic*$^{+/-}$ littermate mice as controls. All mice were housed in a specific pathogen-free animal facility and monitored under an approved animal use protocol approved by the Western University Animal Care Committee.

Cell sorting

Bone marrow (BM) cells were prepared from euthanized mice, and erythrocytes were removed using ammonium-chloride-potassium lysis. Erythrocyte-depleted BM cells were stained with Abs including PE-conjugated anti-CD19 (6D5; BioLegend, San Diego, CA), allophycocyanin-conjugated anti-B220 (RA3-6B2; eBioscience, San Diego, CA), Brilliant Violet 421–conjugated anti-B220 (RA3-6B2; BioLegend), allophycocyanin-conjugated anti-IgM (II/41; BD Biosciences, Franklin Lakes, NJ), FITC-conjugated anti-CD24 (M1/69; BioLegend), PE-conjugated anti-BP-1 (BP-1; BD Biosciences), biotin-conjugated anti-CD43 (S7; BD Biosciences), and PE/cyanine 7–conjugated streptavidin (BioLegend). Infected cultured cells were sorted on the basis of GFP expression. Dead cells were distinguished using SYTOX Blue Dead Cell Stain (Thermo Fisher Scientific, Waltham MA) or 7-aminoactinomycin D viability stain (Thermo Fisher). Cell sorting was performed using a FACSAria II instrument. Flow cytometric analysis was performed using a FACSCanto instrument (BD Immunocytometry Systems, San Jose, CA). Flow data were analyzed using FlowJo 9.1 (BD Biosciences, Ashland, OR). The purity of sorted cells was determined to be ≥98%.

Cell culture

The 38B9 pre-B cell line (30) was cultured in IMDM (Wisent, St.-Jean-Baptiste, QC, Canada) supplemented with 10% FBS (Wisent), 1× penicillin/streptomycin/L-glutamine (Wisent), and 5×10^{-5} M 2-ME (Sigma-Aldrich, St. Louis, MO). Cell lines were maintained in a 5% CO_2 atmosphere at 37°C. To induce differentiation and Igκ V-J rearrangement, 38B9 cells were treated with 0.5 μM imatinib (Sigma-Aldrich) for 48 h (31). PU.1-inducible i660BM cells were cultured as previously described (32) in complete IMDM supplemented with 5% supernatant from the J558L-IL7 cell line (33).

Construction of an Spi-C inducible system

Murine 3xFLAG-tagged Spi-C cDNA (34) was PCR amplified and ligated into the pRetro-Tre3G (Tet-regulatable element third generation) plasmid (Thermo Fisher). MIG-Tet3G was previously described (24). pRetro-Tre3G-Spi-C and MIG-Tet3G retrovirus was produced by transient transfection of Plat-E retroviral packaging cells (35). 38B9 cells were infected by "spinoculation" followed by enrichment for green fluorescent cells using cell sorting. Sorted cells were grown continuously in media supplied with 1 μg/ml puromycin (Bio Basic, Markham, ON, Canada) for 2 wk for selection of resistant cells. The resulting cell line was named "inducible 38B9" (i38B9). To induce Spi-C overexpression, i38B9 cells were treated with doxycycline (Bio Basic) at 1.5 μg/ml.

RT-PCR

RNA was prepared from cultured or sorted cells using the RNeasy kit (Qiagen, Germantown MD), and cDNA was synthesized using the iScript cDNA synthesis kit (Bio-Rad Laboratories, Hercules CA) following the manufacturer's instructions. Reverse transcriptase Quantitative RT-PCR (RT-qPCR) was performed using SensiFast SYBR No-Rox Master Mix (FroggaBio, Toronto, ON, Canada) on a QuantStudio5 instrument (Applied Biosystems/Thermo Fisher Scientific). Relative frequencies of mRNA transcripts were all determined as ratios compared with mRNA levels of β-actin. Genomic DNA was extracted using the Wizard genomic DNA purification kit (Promega, Madison, WI). DNA (500 ng) from each sample was used for PCR amplification of Vκ-Jκ rearrangements (36). For amplification of Igκ Vκ-Jκ5 rearrangements from cDNA, 50-μl PCRs included magnesium chloride at 1.5 mM, Vκ primer at 3.2 μM, Jκ5 primer at 0.2 μM, and 0.5 U of Takara LA-Taq DNA polymerase (Takara Bio, Kusatsu, Japan). The PCR cycle was 95°C for 4 min, followed by 31 cycles at 95°C denaturing for 1 min, 62°C annealing for 1 min, and 72°C extension for 1 min 45 s, and a final 10-min extension at 72°C. Primer sequences are described in Supplemental Table I.

Immunoblotting

Cell lysates were prepared using Nonidet P-40 lysis buffer with moderate agitation for 30 min at 4°C and cleared by centrifugation. Protein concentrations were determined using a bicinchoninic acid protein assay kit (Thermo Fisher Scientific). Lysates were separated using 12% acrylamide gel electrophoresis and transferred to nitrocellulose using semidry transfer (Thermo Fisher Scientific). Blocking was performed using LI-COR Intercept blocking buffer (LI-COR Biosciences, Lincoln, NE). Detection was performed using anti-Flag M2 HRP Ab (Sigma-Aldrich Canada Co., Oakville, ON, Canada) to detect Spi-C or anti-β-actin (C4) Alexa Fluor 680 Ab (Santa Cruz Biotechnology, Dallas, TX).

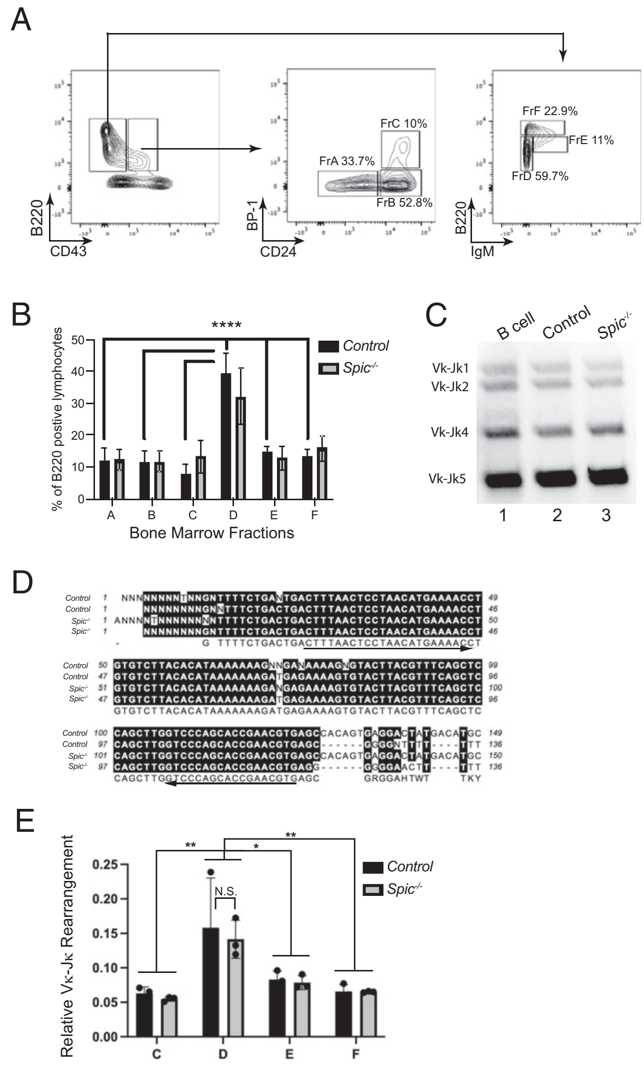

FIGURE 4. Early B cell development is not altered in *Spic*$^{-/-}$ mice. (**A**) Gating strategy for determination of BM developmental stages. Cell surface markers are indicated on the *x*- and *y*-axes, and boxes are labeled with developmental stage identifications (fractions A–F) and representative frequencies within each gate. (**B**) Frequencies of BM B cell fractions A–F (*x*-axis) expressed as a percentage of total BM B220^{+} lymphocytes (*y*-axis). Frequencies of female *Spic*$^{+/+}$ or *Spic*$^{+/-}$ (control) mice and *Spic*$^{-/-}$ F2 mice are indicated in the legend. $^{****}p < 0.0001$ using one-way ANOVA. (**C**) PCR detection of *Igκ* V-J rearrangements in cDNA prepared from fraction D cells enriched from the BM of female F2 mice of the indicated genotype enriched by cell sorting. (**D**) Representative DNA sequences of amplification products of Vκ-Jκ5 rearrangements from two control and two *Spic*$^{-/-}$ mice. Primers indicated by arrows were synthesized on the basis of an aligned sequence and used for quantification of Vκ-Jκ5 rearrangements. (**E**) Quantification of Vκ-Jκ5 rearrangements based on the sequence detected from (D) relative to β-*actin* was carried out by RT-qPCR in BM fractions C–F enriched from female *Spic*$^{+/+}$ and *Spic*$^{+/-}$ (control) and *Spic*$^{-/-}$ F2 mice and is shown on the *y*-axis. N.S., not significant; $^{*}p < 0.05$ and $^{**}p < 0.01$ using one-way ANOVA and Dunnett's multiple comparisons test.

ChIP

Spi-C was induced for 48 h in i38B9 cells with 1.5 μg/ml doxycycline, followed by crosslinking using 1% formaldehyde (MilliporeSigma), and halted using glycine. Crosslinked pellets of up to 1×10^7 cells were flash frozen in liquid nitrogen prior to sonication. Frozen fixed pellets were resuspended in lysis buffer supplemented with HALT protease inhibitor (Thermo Fisher) and sonicated for 25 cycles using the Bioruptor UCD-300 (Diagenode, Sparta, NJ). Immunoprecipitation of 3xFLAG-Spi-C bound chromatin was performed using anti-FLAG M2 magnetic beads (MilliporeSigma). DNA was eluted from input chromatin or immunoprecipitated chromatin by heating. Eluted DNA was purified using the ChIP DNA clean and concentrator kit (Zymo Research, Irvine CA). qPCR analysis was performed using primers described in Supplemental Table I.

Statistical analysis

Statistical analysis was performed using Prism 9.4.1 (GraphPad Software, La Jolla, CA). Data are presented as mean ± SEM unless otherwise indicated. Statistical significance was determined using tests indicated in the figure legends. All data points in the figures in this study represent independent biological replicate experiments.

Results

Induced expression of Spi-C inhibits Igκ *rearrangement in an inducible cell line*

To study the role of Spi-C in *Igκ* transcription and rearrangement, we used the Abelson (Abl) kinase-transformed pre-B cell line 38B9 (30). Cell cycle arrest, *Igκ* V-J rearrangement, and B cell differentiation can be induced in 38B9 cells using the Abl kinase inhibitor imatinib (also known as STI571 or Gleevec [Novartis]) (31). We constructed a two-vector doxycycline-inducible system for ectopic expression of 3xFLAG-tagged murine Spi-C in 38B9 cells (Fig. 1A). With this system, Spi-C expression could be induced in inducible 38B9 (i38B9) cells up to 10-fold at 48 h of doxycycline treatment at a concentration of 1500 ng/ml (Fig. 1B). Induction of Spi-C did not affect cell cycle progression or induce *Igκ* recombination in i38B9 cells (Supplemental Figs. 1 and 2).

Next, 38B9 cells or i38B9 cells were treated with imatinib to induce *Igκ* V-J rearrangement (Fig. 1C). Imatinib treatment induced 38B9 cell cycle arrest by 48 h, as shown by flow cytometric analysis of a

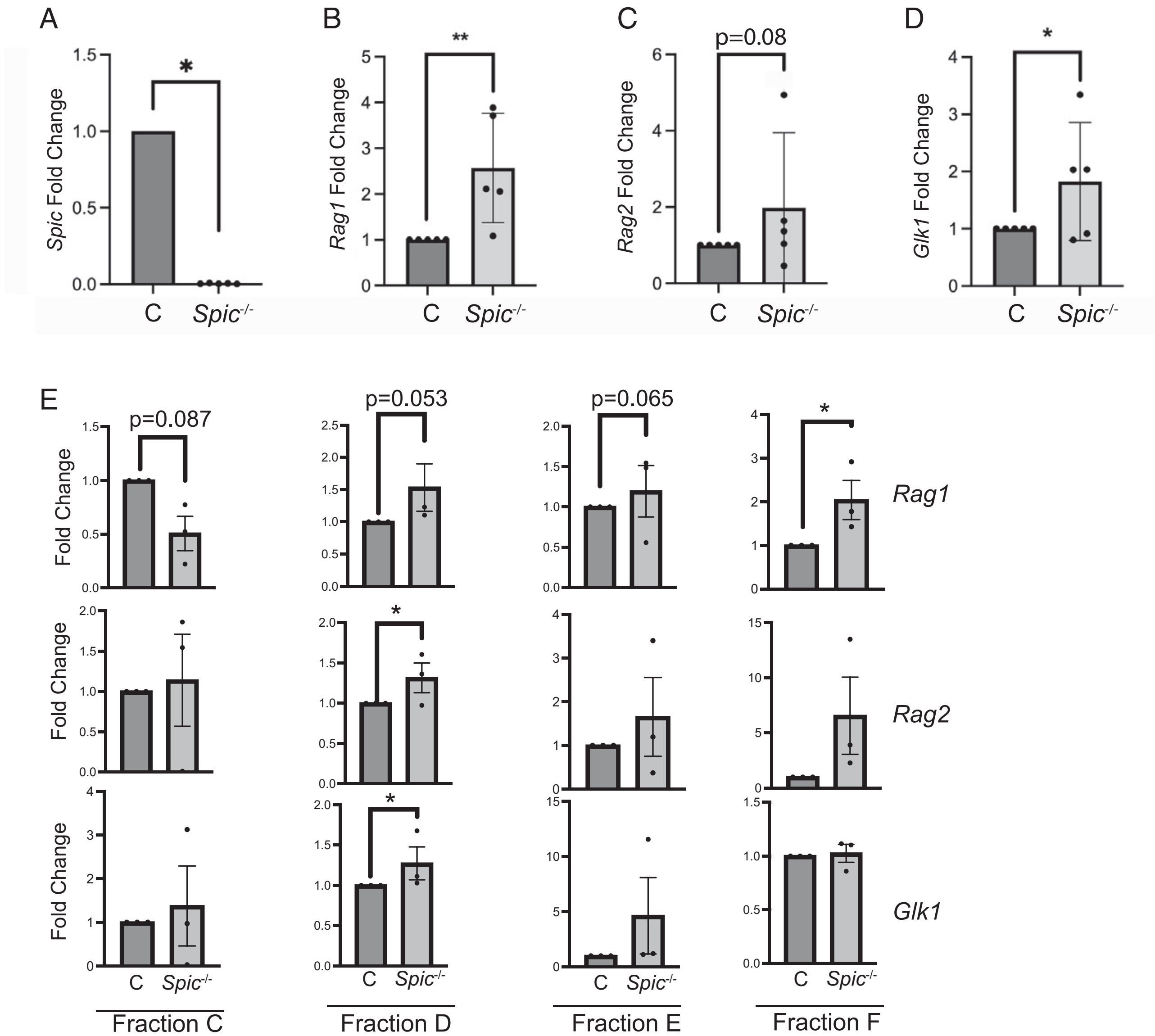

FIGURE 5. Increased relative levels of *Rag1* and *Glk1* transcripts in BM B cells from *Spic*$^{-/-}$ mice. (**A–D**) RNA was prepared from B220$^+$ CD19$^+$ B cells enriched from the BM of female F2 *Spic*$^{-/-}$ or control mice. RT-qPCR was performed for the genes indicated on the *y*-axis. Relative expression (gene/β-*actin* ratios) is indicated on the *y*-axis. *$p < 0.05$, **$p < 0.01$ using one-sample *t* and Wilcoxon tests. (**E**) RNA was prepared from BM fractions C–F enriched from the BM of female F2 *Spic*$^{-/-}$ or control mice. RT-qPCR was performed for the genes indicated on the right side. Expression relative to control is indicated on the *y*-axis. *$p < 0.05$ using one-sample *t* and Wilcoxon tests.

shift from high forward scatter to low forward light scatter, suggesting a reduction in cell size (Supplemental Fig. 1A). *Spic* was expressed at low levels in untreated 38B9 or i38B9 cells and was increased substantially by 0.5 μM imatinib treatment alone (Fig. 1D, 1E). Prior to imatinib treatment, both 38B9 and i38B9 cells had detectable V_k-J_k1 and V_k-J_k2 rearrangements but undetectable Vκ-Jκ4 and Vκ-Jκ5 rearrangements (Supplemental Fig. 2, Fig. 1F). Imatinib treatment for 24 h induced Vκ-Jκ4 rearrangements in both 38B9 and i38B9 cells (Supplemental Fig. 2B).

Next, the combined effects of inducing Spi-C using doxycycline and inducing *Igκ* rearrangement with imatinib were investigated. *Spic* mRNA levels were induced over 50-fold by combined doxycycline and imatinib treatment (Fig. 1D, right side). Strong induction of Spi-C protein by doxycycline and imatinib treatment was confirmed by anti-FLAG immunoblotting (Fig. 1E). Induction of Spi-C in i38B9 cells inhibited imatinib-induced *Ig* Vκ-Jκ rearrangement at the 48-h time point (Fig. 1F, right panels). This result suggests that ectopic expression of Spi-C negatively regulates *Igκ* rearrangement in pre-B cells.

Repression of Igκ *germline and* Rag1 *transcripts by Spi-C*

Igκ rearrangement might be inhibited by Spi-C through repression of *Igκ* transcription, repression of RAG1/2 expression, or a combination of both. To determine the mechanism of inhibition of *Igκ* rearrangement by Spi-C, gene expression was examined using RT-qPCR. *Vcam1* encodes vascular cell adhesion protein-1 and was previously shown to be directly activated by Spi-C (20). *Rag1* and *Rag2* encode the recombinase-activating 1 and 2 proteins, whereas *Glk1* encodes an *Igκ* germline "sterile" transcript that correlates with locus accessibility (Fig. 2A) (6). *Vcam1* was activated following imatinib treatment in the presence or absence of Dox (Fig. 2B). *Glk1* but not *Rag1* or *Rag2* transcripts were inhibited by Spi-C at the 24-h time point (Fig. 2C–2E). However, *Rag1*, *Rag2*, and *Glk1* mRNA transcript levels were all reduced at 48 h after imatinib treatment in doxycycline-induced i38B9 cells (Fig. 2C–2E). Therefore, ectopic expression of Spi-C inhibits *Igκ* germline transcription, *Rag1*, and *Rag2* transcription in 38B9 cells.

Early B cell development is not altered in Spic$^{-/-}$ *mice*

We set out to determine the effect of *Spic* loss of function on gene expression in developing pre-B cells in mice. *Spic*$^{-/-}$ mice are born at low frequency due to preimplantation lethality of undetermined cause (20, 37). We found that *Spic*$^{-/-}$ mice on a C57BL/6 strain background were rarely generated, either at 3 wk of age or fetal 14.5 d postimplantation (Table I). However, by intercrossing F1 progeny of C57BL/6 *Spic*$^{+/-}$ × 129Sv mice, we were able to obtain *Spic*$^{-/-}$ mice with a frequency of 14.8% by 3 wk of age (Table I, Fig. 3A). This outcome suggests that *Spic*$^{-/-}$ mice on a mixed C57BL/6 and 129.Sv strain background have reduced preimplantation lethality. This procedure allowed generation of *Spic*$^{-/-}$ mice for further analysis.

There are substantial sex differences between frequencies of lymphocytes in various mouse strains (38). We found that female B6/129 F2 mice had increased frequencies of CD19$^+$ B220$^+$ B cells in the BM compared with male mice (Fig. 3B, 3C). There were no significant differences in total BM cell numbers between *Spic*$^{-/-}$ and control mice (Fig. 3D, 3F). However, *Spic*$^{-/-}$ mice had higher total numbers of CD19$^+$ B220$^+$ B cells in female mice compared with control animals (Fig. 3E). There were no significant differences between *Spic*$^{-/-}$ and control mouse B cell frequencies in male mice (Fig. 3G). This result suggests a sex-specific effect of Spi-C on B cell development.

The frequency of BM B cell development stages was determined using the Hardy scheme in which B cell development is separated into fractions A–F (39). No significant differences were detected between *Spic*$^{-/-}$ female mice and *Spic*$^{+/+}$ or *Spic*$^{+/-}$ female control mice (Fig. 4A, 4B) or between male mice of the same genotypes (data not shown). Next, *Igκ* V-J rearrangements were determined. RT-PCR analysis of RNA prepared from enriched fraction D BM pre-B cells showed similar frequencies of Vκ-Jκ rearrangements in *Spic*$^{-/-}$ mice and control mice (Fig. 4C). To perform quantitative analysis, Vκ-Jκ5 PCR products were sequenced and used to design qPCR primers (Fig. 4D). RT-qPCR analysis of Vκ-Jκ5 rearrangements revealed that mRNA levels were higher in fraction D pre-B cells than in fraction C, E, or F (Fig. 4E). However, no difference was found between *Spic*$^{-/-}$ mice and control mice (Fig. 4E). In summary, these results suggest that loss of Spi-C function does not impair early B cell development.

Deletion of the Spic *gene in mice leads to elevated levels of* Rag1 *and* Igκ *sterile transcripts*

The low frequency of BM B cells in male B6/129 F2 mice *Spic*$^{-/-}$ mice made enrichment of B cells from male mice by cell sorting technically unfeasible. We therefore enriched total BM CD19$^+$ B220$^+$ B cells from female B6/129 F2 *Spic*$^{-/-}$ mice, as well as female littermate *Spic*$^{+/+}$ or *Spic*$^{+/-}$ mice as controls. RNA was prepared from enriched B cells for RT-qPCR analysis of gene expression. *Spic* mRNA transcripts were undetectable in *Spic*$^{-/-}$ BM B cells, as expected (Fig. 5A). *Rag1* and *Glk1* were expressed at significantly higher levels in B cells from *Spic*$^{-/-}$ female mice than those from *Spic*$^{+/+}$ or *Spic*$^{+/-}$ female mice (Fig. 5B, 5D). *Rag2* trended toward higher levels of expression in *Spic*$^{-/-}$ female

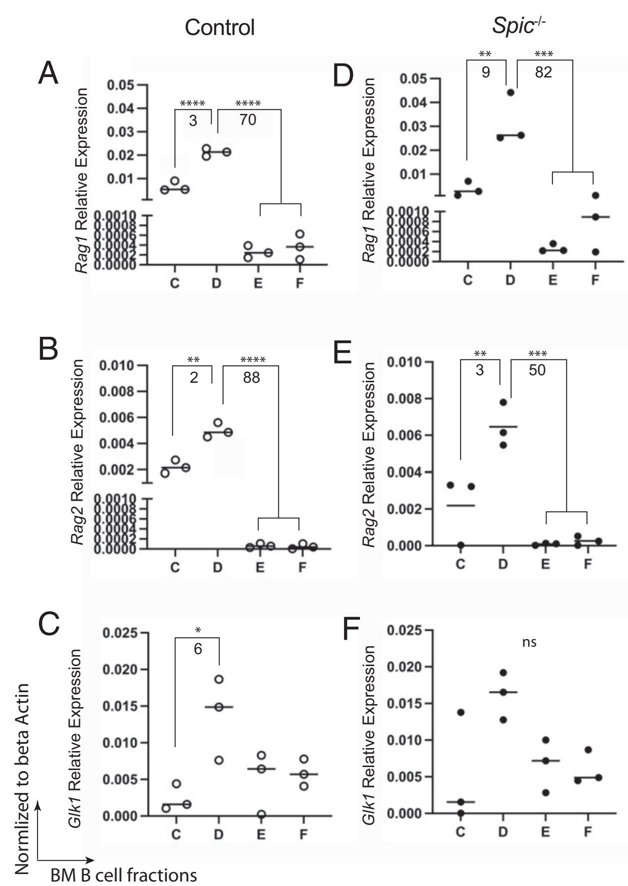

FIGURE 6. Analysis of *Rag1*, *Rag2*, and *Glk1* mRNA transcript levels in fractions C–F BM B cells from *Spic*$^{-/-}$ mice. For (**A–F**), RNA was prepared from BM fractions C–F enriched from the BM of female F2 *Spic*$^{-/-}$ or control mice. RT-qPCR was performed for the genes indicated on the *y*-axis in each panel. Relative expression (gene/β-*actin* ratios) is indicated on the *y*-axis. Numbers indicate fold difference between groups. *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$, ****$p < 0.0001$, using one-way ANOVA.

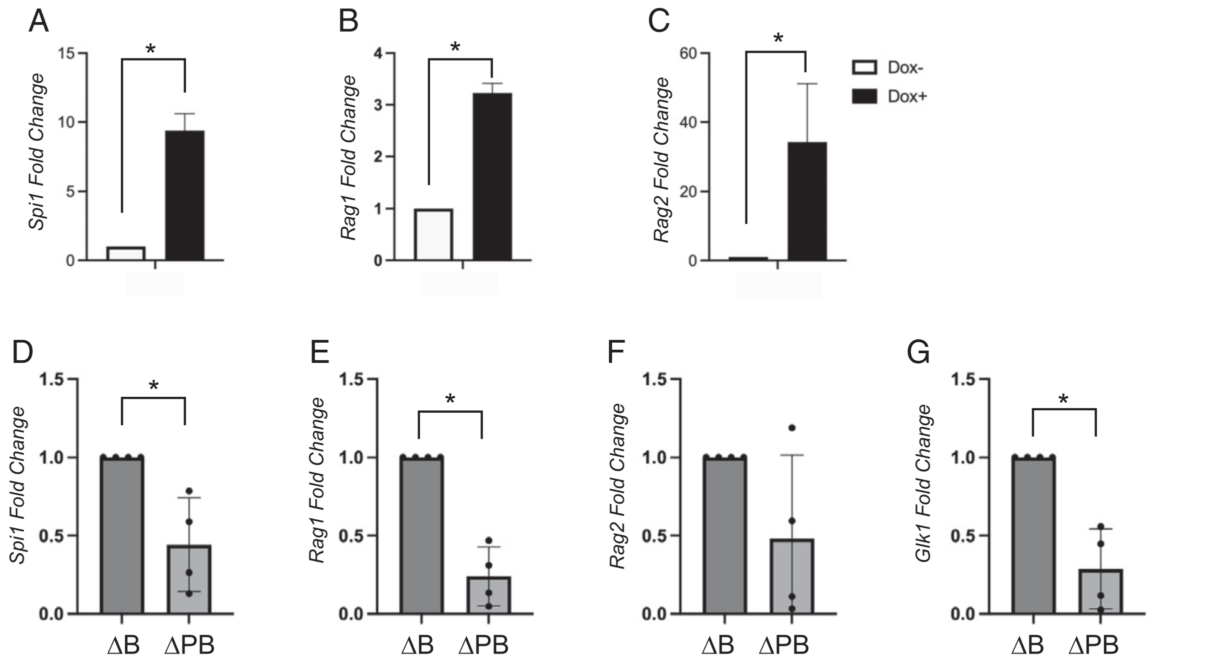

FIGURE 7. PU.1 activates *Ig*κ germline and *Rag1* transcription. (**A–C**) PU.1-inducible i660BM cells were treated with 70 ng/ml doxycycline (Dox^+) or without (Dox^-) for 48 h, followed by preparation of RNA. RT-qPCR was performed to determine relative expression levels of the genes indicated on the *y*-axis normalized to β-actin levels. $*p < 0.05$ using one-sample *t* and Wilcoxon tests. (**D–G**) PU.1 is required for gene expression in fraction D pre-B cells. Fraction D pre-B cells were enriched by cell sorting from BM of female Mb1-CreΔPB mice lacking both PU.1 and Spi-B (ΔPB) or $Spib^{-/-}$ mice as a control (ΔB). RT-qPCR was performed to determine relative levels of the genes indicated on the *y*-axis. $*p < 0.05$ using one sample *t* and Wilcoxon tests.

mice than in $Spic^{+/+}$ or $Spic^{+/-}$ female mice (Fig. 5C). These results suggested that Spi-C controls levels of *Rag1* and *Glk1* transcripts in BM B cells.

Next, fractions C, D, E, and F cells were enriched by cell sorting from $Spic^{-/-}$ female mice and $Spic^{+/+}$ or $Spic^{+/-}$ female mice, and RNA was prepared to determine gene expression using RT-qPCR. *Rag1* was expressed at higher levels in $Spic^{-/-}$ fraction D cells and fraction F cells than in controls (Fig. 5E). *Rag2* and *Glk1* were expressed at higher levels in $Spic^{-/-}$ fraction D cells than in control fraction D cells. The majority of fraction D cells represent pre-B cells that have stopped proliferating in response to IL-7 and have initiated rearrangement of Igκ but do not yet express a BCR (39). In agreement with this observation, we found that fraction D cells expressed higher levels of *Rag* and *Glk1* mRNA transcripts than fraction C, E, or F cells (Fig. 6). For *Rag1*, there was a larger difference between fractions C and D in $Spic^{-/-}$ than in wild type (9-fold compared with 3-fold; Fig. 6A and 6D) or between fractions E/F to D (82-fold compared with 70-fold; Fig. 6A and 6D). In summary, our results suggest that Spi-C represses *Rag* and *Glk1* transcription starting at the pre-B (fraction D) cell stage.

PU.1 activates Igκ *germline and* Rag1 *transcription*

PU.1 and Spi-B are highly related to Spi-C in the ETS transcription factor family. We previously showed that PU.1 is an activator of *Ig*κ transcription and rearrangement during B cell development (24). To further explore this observation, we made use of the i660BM pre-B cell line in which PU.1 is doxycycline inducible (32). Doxycycline induction of PU.1 in i660BM cells for 48 h (Fig. 7A) increased *Rag1* and *Rag2* mRNA transcript levels (Fig 7B, 7C). This result suggests that PU.1 is an activator of *Rag* transcription. Next, we looked at mice in which PU.1 is inducibly deleted under control of the *Mb1* (*Cd79a*) gene starting at the pro-B stage (Mb1-CreΔPB mice) (24). These mice are also germline null for the *Spib* gene. We found that *Rag1* and *Glk1* mRNA transcripts were decreased in fraction D cells enriched from Mb1-CreΔPB mice compared with $Spib^{-/-}$ control mice (Fig. 7D–7G). These data suggest that PU.1 functions as an activator of *Ig*κ and *Rag1* transcription and *Ig*κ rearrangement. Taken together with the previous results, these data suggest that Spi-C and PU.1 act in an opposite manner to regulate *Ig*κ and *Rag1* transcription and *Ig*κ rearrangement starting at the small pre-B cell stage.

Direct interaction of PU.1 and Spi-C with a site in the Rag *locus*

To determine if Spi-C regulates *Rag* transcription directly by interaction with a binding site in a regulatory region, we performed anti-Spi-C ChIP. 3xFLAG-Spi-C was induced by doxycycline in i38B9 cells for 48 h, after which cells were formaldehyde fixed and chromatin prepared for immunoprecipitation. ChIP was performed using anti-FLAG Ab as described in *Materials and Methods*. First, qPCR analysis was performed on a previously identified Spi-C binding site (23) in the first intron of the *Bach2* gene (Fig. 8A). Using qPCR analysis, Spi-C binding at *Bach2* region of interest 1 was increased 40-fold compared with binding at a negative control region (Fig. 8B). Next, qPCR analysis was performed to determine Spi-C binding at a previously identified PU.1 binding site within the *Rag* locus (24) (Fig. 8C). Four independent biological replicate experiments found 8–45-fold increased binding of Spi-C at the *Rag* peak compared with a negative control region in the same locus (Fig. 8D–8G). In summary, Spi-C and PU.1 interact with a binding site located directly upstream of the *Rag1* gene, suggesting that Spi-C and PU.1 regulate Rag transcription directly.

Discussion

In this study, we investigated the ability of the ETS transcription factor Spi-C to negatively regulate *Ig*κ L chain recombination. Using an inducible expression system in a pre-B cell line, we found that Spi-C negatively regulated *Ig*κ rearrangement, *Ig*κ transcript levels, and *Rag1* transcript levels. By analyzing BM-derived B cells from $Spic^{-/-}$ mice, we found that *Ig*κ and *Rag1* transcript levels were negatively regulated by Spi-C in small pre-B cells. In contrast, we found that *Ig*κ and *Rag1* transcript levels were activated by PU.1 in small

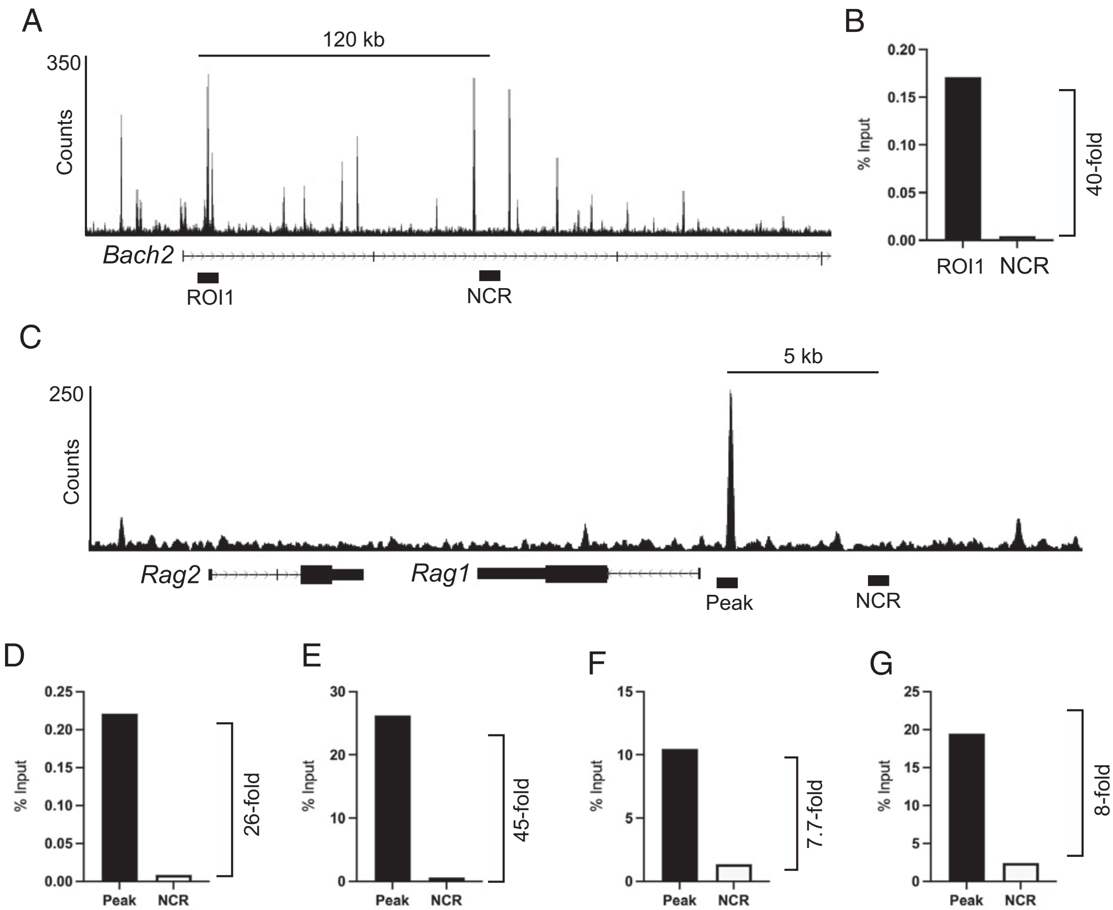

FIGURE 8. Direct interaction of PU.1 and Spi-C with a site in the *Rag* locus. (**A**) University of California, Santa Cruz genome browser track of PU.1 ChIP-sequencing data showing the *Bach2* gene. Black bars indicate region of interest 1 (ROI1) and negative control region (NCR). (**B**) Enrichment of Spi-C at ROI1 relative to the NCR. Anti-Spi-C ChIP was performed in i38B9 pre-B cells as described in *Materials and Methods* and is shown as a percentage of input (*y*-axis). (**C**) University of California, Santa Cruz genome browser track of PU.1 ChIP-sequencing data in i660 pre-B cells showing the *Rag* locus. Black bars indicate a region of PU.1 binding (Peak) and a negative control region (NCR). (**D**–**G**) Enrichment of Spi-C at the PU.1 binding site (Peak) relative to NCR. (D–G) represent four biological replicates. Fold enrichment is shown as a percentage of input.

pre-B cells. Using ChIP analysis, we identified a binding site for PU.1 and Spi-C located 5′ of the *Rag1* promoter. We conclude that Spi-C and PU.1 counterregulate *Ig*κ L chain recombination and *Rag1* transcription in small pre-B cells.

DNA DSBs induced at the *Ig*κ L chain locus in small pre-B cells have been demonstrated to negatively feed back on *Rag* and *Ig*κ transcription to promote allelic exclusion at the *Ig*κ locus (16). The signaling pathway induced by DSBs to promote allelic exclusion requires ATM kinase and the Nemo protein (16). This signaling pathway activates several genes, including *Spic* (18). Spi-C is highly related to PU.1 and Spi-B and can interact with and/or compete for similar binding sites in the genome of B cells and macrophages (23, 29, 40). Spi-C can repress *Ig*κ recombination and cell division by interacting with BCLAF1 to displace PU.1, resulting in repression of the *Syk*, *Blnk*, and *Ig*κ genes (18, 19) Although *Rag1* transcription was shown to be negatively regulated by DSBs, it has not been clear what factors mediate this transcriptional repression (41). Our study now demonstrates that Spi-C is a direct negative regulator of *Rag1* transcription in pre-B cells.

PU.1 has been shown in multiple studies to function primarily as an activator of transcription (42). In addition, PU.1 functions as a pioneer protein, having the ability to interact with binding sites in closed chromatin and to recruit histone acetyltransferases to open chromatin sites (43). In our previous studies and in the study by Soodgupta et al. (19), Spi-C appeared to function as a negative regulator of gene transcription primarily by displacing PU.1 from binding sites. ChIP studies suggest that most Spi-C binding sites are also PU.1 binding sites, whereas there are fewer unique Spi-C binding sites (23). It remains to be determined whether Spi-C interaction with target sites requires prior PU.1 binding or pioneer activity.

The *Ig*κ 3′ enhancer contains an important ETS transcription factor binding site that regulates B cell versus T cell specificity of Vκ-Jκ joining (7). PU.1 was shown to interact with this site to promote *Ig*κ transcription and accessibility (44, 45). Because of functional redundancy and complementarity with PU.1, Spi-B very likely also regulates *Ig*κ transcription through this site (24). Spi-C was shown to displace PU.1 interaction with binding sites in the *Ig*κ locus (19). Therefore, Spi-C likely negatively regulates the *Glk1* mRNA transcript by direct interaction with PU.1 interaction sites in the *Ig*κ locus.

In all experiments performed in this study, *Rag1* was affected to a greater degree than *Rag2* by Spi-C ectopic expression or deletion. We speculate that this is because of the proximity of the *Rag1* transcription start site to the binding site that we identified for PU.1/Spi-C. Mechanisms of transcriptional regulation of the *Rag* locus have not been well studied, with only the identification of the ERag enhancer reported for many years (14). However, a more detailed study of *Rag* transcriptional regulation was reported in 2020 by Miyazaki et al. using assay for transposase-accessible chromatin with sequencing to identify additional regulatory regions named R-TEn, R2B, R1pro, and R1B (15). R1pro represents the *Rag1* promoter region and was shown to interact with multiple transcription factors, including E2A, Pax5, ETS1, Ikaros, and IRF4. We have now shown that Spi-C and PU.1 also interact with the R1pro region, adding to the evidence that this is an important regulatory region for the *Rag1* gene.

We speculate that in small B cells, PU.1 activates *Igκ* and *Rag* transcription using its pioneer activity. PU.1 is a strong transcriptional activator of *Igκ* (24) and, as we show here, *Rag1* to promote *Igκ* recombination. Following RAG1/2-mediated DNA cleavage at one *Igκ* allele, The *Spic* gene is activated, resulting in Spi-C interaction with BCLAF1 to displace PU.1 from target genes in small pre-B cells. Spi-C expression may persist into the transitional 1 stage to modulate BCR signaling. In this manner, PU.1 and Spi-C are proposed to function as counterregulators of *Igκ* allelic exclusion and are potentially involved in regulation of BCR signaling and receptor editing. In summary, our experiments support the idea that Spi-C and PU.1 counterregulate *Igκ* transcription and *Rag1* transcription to effect *Igκ* recombination in small pre-B cells.

Acknowledgments

We thank Dr. Malay Haldar (Perelman School of Medicine, University of Pennsylvania) for advice regarding breeding and genotyping of *Spic*$^{-/-}$ mice. We thank Kristin Chadwick and the London Regional Flow Cytometry Facility for assistance with cell sorting.

Disclosures

The authors have no financial conflicts of interest.

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